



Meet the Partner Session

National AI Student Challenge 2026

6 March 2026





Challenge Deep Dive

Low Hon Zheng, Senior AI Engineer
AI Architecture & Solution



Meet the Singtel AIDA Team!



Hou Liang

AD, Business Partners
& Enablement



Hon Zheng Low

Senior AI Engineer



Cecilia Lin

Senior Data Scientist



Yiting Jin

Senior Data Scientist



The Problem: AI in a Changing World



AI in a Changing Data World

The Reality of Model Drift:

Data Evolves: The world is non-static; consumer behaviour, market trends, and environmental factors change daily.

The Stability Fallacy: Models inherently assume that the future will mirror the distribution of the training data.

Silent Degradation: Unlike code that crashes, AI models often continue to run, quietly becoming less accurate overtime.

GIGO Effect: Garbage In, Garbage Out. Flawed or stale data inputs inevitably result in flawed, untrustworthy intelligence.

Real World Case Study



- **The Model:** Predicted home values to automate "flipping" houses.
- **The Drift:** COVID-19 caused labor and material costs to spike unpredictably.
- **The Failure:** The AI kept buying high while the market cooled (Silent Degradation).
- **The Cost:** **\$500M+ loss** and a 25% workforce reduction.

An AI model is not a 'set-it-and-forget-it' asset. It is a perishable product that requires continuous monitoring, retraining, and validation.

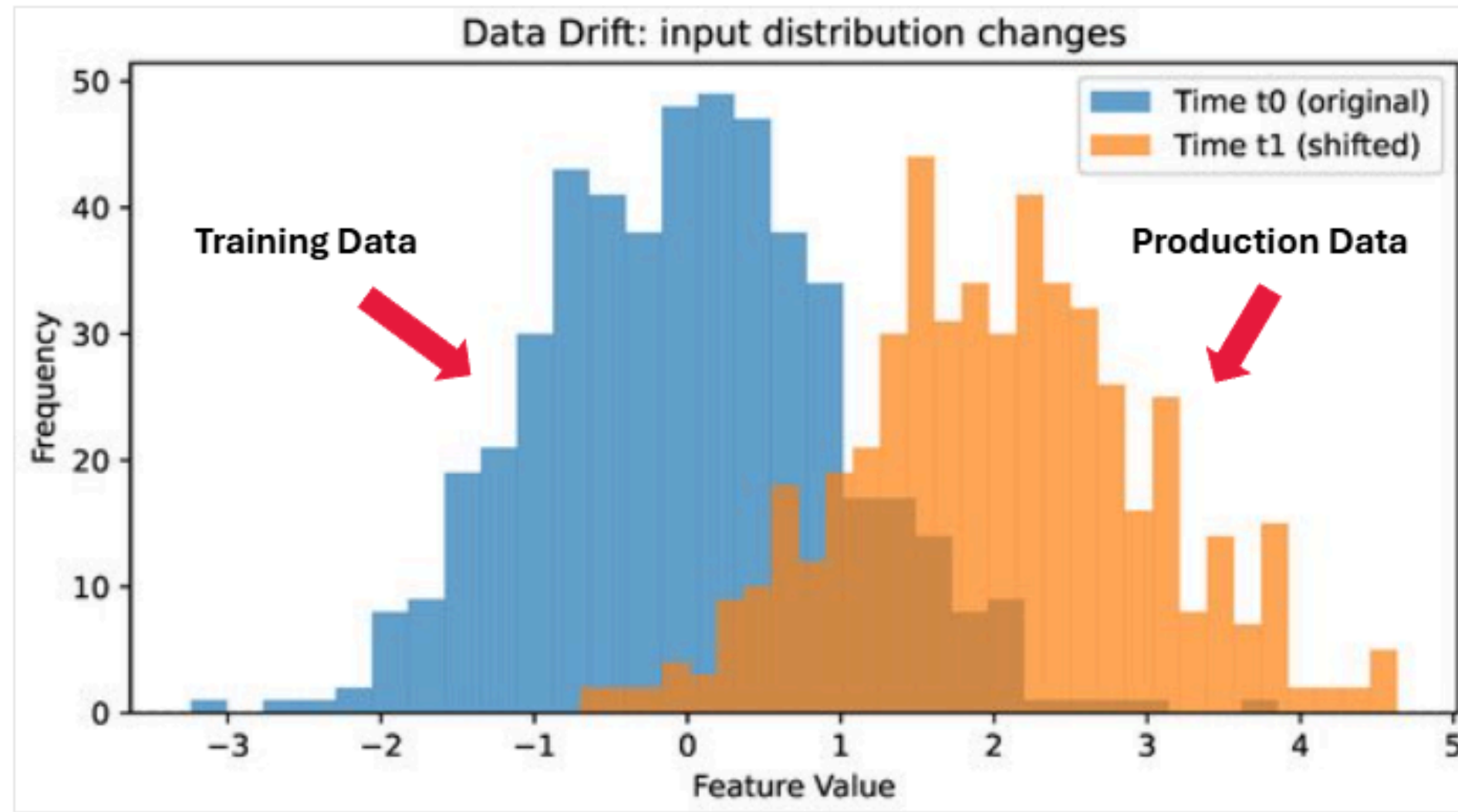


Understanding Data Drift

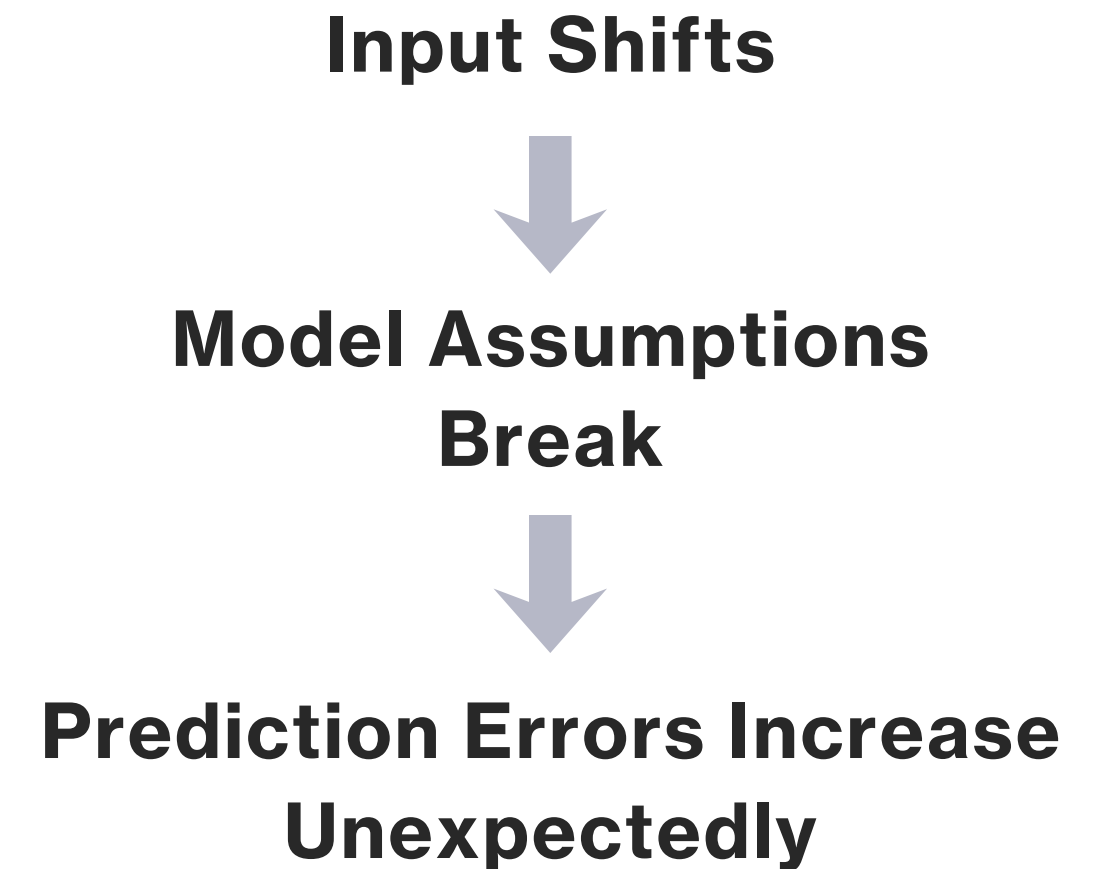


What Is Data Drift – And Why It Matters

Data Drift happens when the data your model sees in production no longer looks like the data it was trained on



The model learned the blue distribution – but now it sees orange



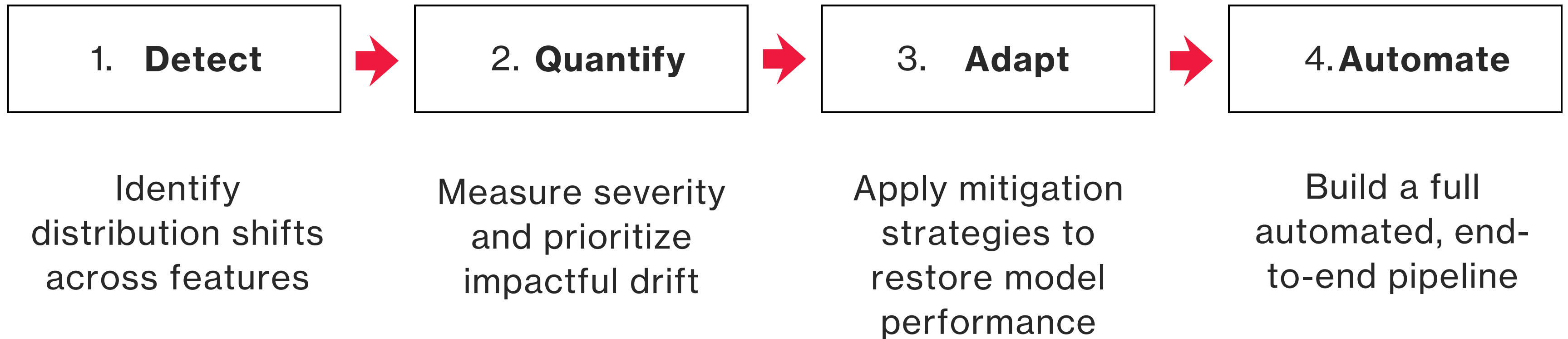


Challenge Requirement



The Challenge: Build Adaptive, Resilient AI

From Drift Detection to Model Adaptation



Detecting and Quantifying Data Drift

Step 1: Identify Feature Types

- Numerical
- Categorical
- High cardinality / Sparse
- Time-based features

Step 2: Select Appropriate Techniques

- Kolmogorov-Smirnov (K-S) test
- Population Stability Index (PSI)
- Chi-Squared test
- Drift Classifiers

Step 3: Measure Drift Severity

- Statistical significance (p-value)
- Effect size / distance score
- Threshold-based flagging
- Feature-level drift scoring

Technical Constraints & Standardized Environment

Model Standardization

LightGBM (v4.6.0)

"verbosity": -1

"objective": "binary"

"is_unbalance": True

"random_state": 42

"importance_type": 'gain'

Runtime Environment

Python v3.12

CPU only

Purpose of Constraints

Ensure fair comparison

Focus evaluation on drift handling

Prevent overfitting via complex models

Adaptive Drift Mitigation

Solving Drift through Data Engineering (Fixed Hyperparameters)

Recency

- Sliding Window
 - Train on most recent N days
- Weighted Decay
 - Older data -> Lower importance
- Seasonality Matching
 - Use statistically similar historical blocks

Robustness

- Invariant Transformations
 - Log / Robust scaling
- Delta-Based Features
- Binning / Discretization

Monitoring

- Input Re-Alignment
 - Centering the data back to training norms before model inference
- Drift-Based Pruning
 - Dropping features with high drift

Automated End-to-End Pipeline

1. Standardized Entry Point

```
pip install -r requirements.txt

python ./src/main.py --train_data_filepath <train_data_filepath> --test_data_filepath <test_data_filepath>
```

2. Required Outputs

a. Data Drift Detection and Mitigation

Columns with Drift	Column Type	Drift Description	Drift Mitigation
Col A	Int	Feature demonstrates greater left-skewness in test set compared to training set	Seasonality Matching
Col B	Object	Feature has new set of categorical features in test set compared to training set	Drop Feature
Col D	Float	Feature ranges explode in test set	Feature Scaling

Output: Test set predicted probabilities

CustomerID	probability_score
b677211fb5b146e6	0.843
ea87c0609e1e49a9	0.600
51c2c0949fc549c9	0.481

b. Data Drift Detection and Mitigation Time Taken

Time Taken (s)
21.3

c. Model Performance

	AU-PRC
Train Set	0.714
Test Set	0.725

Dataset Overview

Public Dataset (Binary Classification)

- Primary Key: CustomerID
- Time Column: Month
- Target Column: ChurnStatus

Train Data

- 70,430 rows
- 45 columns
- Mixed features types

Test Data

- 14,086 rows
- 45 columns
- Mixed features types

Hidden Dataset (Binary Classification)

- Limited to < 10 million rows
- Limited to < 500 features
- Mixed feature types
- Feature names may differ
- Same Primary Key, Time and Target column as Public Dataset
- Used for final scoring



Evaluation Criteria



How Your Solution Will Be Evaluated

Performance (70%)

Drift Detection (35%)

Precision of detected drift features

Ability to detect multiple drift signals

Appropriate metric selection

Drift Mitigation (35%)

Hidden Test Set AU-PRC

Stability of performance

Robustness & Scalability (15%)

Script runnability

main.py runs without errors and manual intervention

Works on unseen datasets

Runtime efficiency (< 10 mins runtime for drift detection and drift mitigation)

Report (15%)

Novelty of solution

Novelty of approach

Clear explanation of detection and mitigation strategy

Comparison with alternative approaches to arrive at the final solution

Discussion of limitations

Dashboard (Bonus)

Clear visualization of drift severity

Interactive exploration capability

Ease of deployment

Presentation (For Top 10 finalists)

Technical clarity

Depth of understanding

Communication quality

Professional delivery



Timeline for Submission



Submissions

6 March 2026

Challenge Begins

- Files will be available in GitHub
- Questions can be raised in the discussion tab of the repo



12PM 17 April 2026

Submissions Due

- Files required for submission
 - Clean folder structure with main.py in src/ directory
 - README file, Public Test Set prediction .csv file, PDF Report, Model joblib file, requirements.txt
- Submit Microsoft Form + add us as collaborators

5 May 2026

Finalists Announced

- Finalists to prepare for presentation in Grand Finals



22/23 May 2026

Grand Finals

- Top 3 teams to receive prizes

Contact Us

GitHub Repo with Challenge Materials and Discussion Forum



GOOD LUCK, STAY CURIOUS AND HAVE FUN :”)